

SEMESTER VII · INFORMATION TECHNOLOGY · MAJOR PROJECT

Spendly

Offline-First Zero-Knowledge Peer-to-Peer Finance Ecosystem

A privacy-preserving, serverless financial infrastructure — operating entirely offline, secured through zero-knowledge cryptography, and built for the next generation of commerce.

Student Name · Enrollment Number · Semester VII · Information Technology · College Name



The Problem with Modern Finance

Every digital transaction today creates a vulnerability — a traceable record stored on someone else's server, exposed to breaches, surveillance, and dependency. The current architecture is fundamentally broken.

Privacy at Risk

Centralized databases are prime targets. Every transaction leaks behavioral, financial, and personal data to third parties.

Internet Dependency

POS systems, billing apps, and wallets fail without connectivity — crippling commerce in low-bandwidth or remote environments.

Monthly POS Costs

Merchants pay recurring subscription fees for hardware and software that could operate independently at zero cost.

Data Breach Exposure

A single centralized breach compromises millions of users simultaneously. There is no containment, no recovery.





Introducing Spendly

Three interconnected applications forming a complete, privacy-first financial ecosystem — no servers, no subscriptions, no compromise.

Spendly

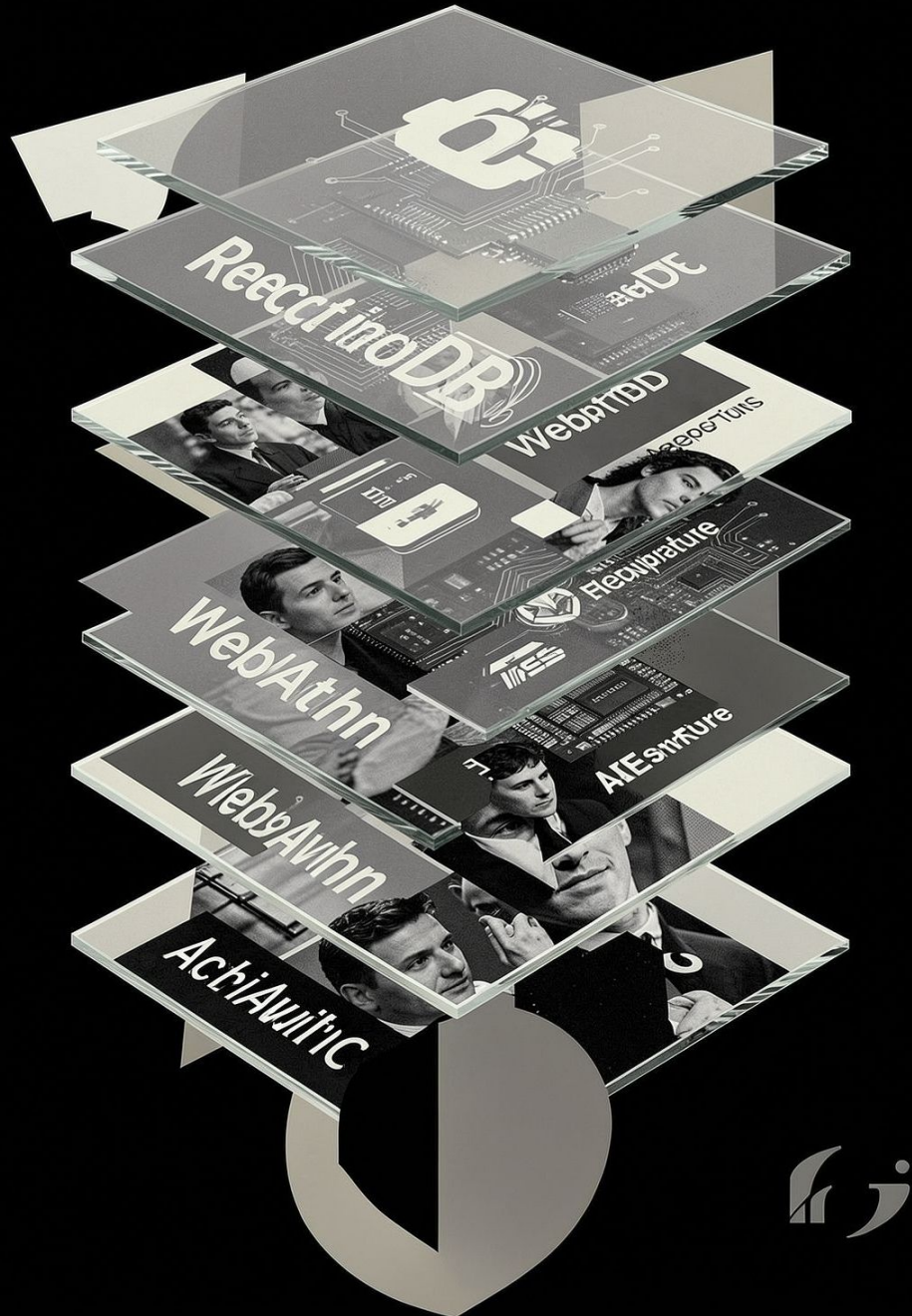
The consumer-facing app. Track expenses, claim offers, scan barcodes, and manage personal finance — entirely offline, entirely private.

Spendly Shop

A merchant POS application. Generate invoices, manage inventory, accept payments, and sync peer-to-peer — zero monthly fees.

Spendly Radar

The discovery layer. Broadcast localized offers and coupons via WebRTC mesh — reaching nearby users without internet or central servers.



System Architecture

Spendly's architecture is a deliberate departure from cloud-native design. Every layer is chosen to enable offline operation, peer-to-peer communication, and cryptographic security — without a single centralized server in the critical path.

1

Frontend Layer

React + PWA — a fully installable, offline-capable progressive web app with responsive UI and IndexedDB local storage via Dexie.js.

2

Networking Layer

WebRTC + PeerJS — direct browser-to-browser encrypted channels for real-time peer-to-peer data exchange without intermediaries.

3

Edge Layer

Cloudflare Workers + KV — optional lightweight edge functions for non-sensitive metadata, never for financial data or user state.

4

Security Layer

WebAuthn + AES-256-GCM — hardware-backed biometric authentication and authenticated encryption for all stored and transmitted data.

How Spendly Works

From discovery to settlement — every step is local, encrypted, and peer-to-peer. No round-trips to a server. No latency. No exposure.

01

Offer Discovery

Spendly Radar broadcasts localized coupons via WebRTC mesh to nearby devices.

02

Coupon Claim

User claims offer — stored locally in IndexedDB, validated offline.

03

Barcode Scan

Product identified via on-device scan — no external API call required.

04

Product Lookup

Inventory matched from local cache synchronized peer-to-peer with Spendly Shop.

05

Offer Validation

Coupon applied and validated through encrypted local logic — zero server contact.

06

Invoice Generation

Spendly Shop generates signed invoice, transmitted directly to user's device via WebRTC.

07

Invoice Sharing

Encrypted invoice shared peer-to-peer — user stores locally, merchant retains copy.

08

Expense Saving & Analytics

Spending categorized and analyzed entirely on-device — insights never leave the user's hardware.

Technology Stack

Every technology in Spendly is selected for a single reason: enabling a fully functional, secure, and private financial system without a traditional backend.

Frontend

React · PWA · Dexie.js

Responsive, installable, and fully offline-capable user interface with structured local data management.

Networking

WebRTC · PeerJS

Direct encrypted peer-to-peer channels — no relay servers, no intermediaries, no latency.

Database

IndexedDB · Cloudflare KV

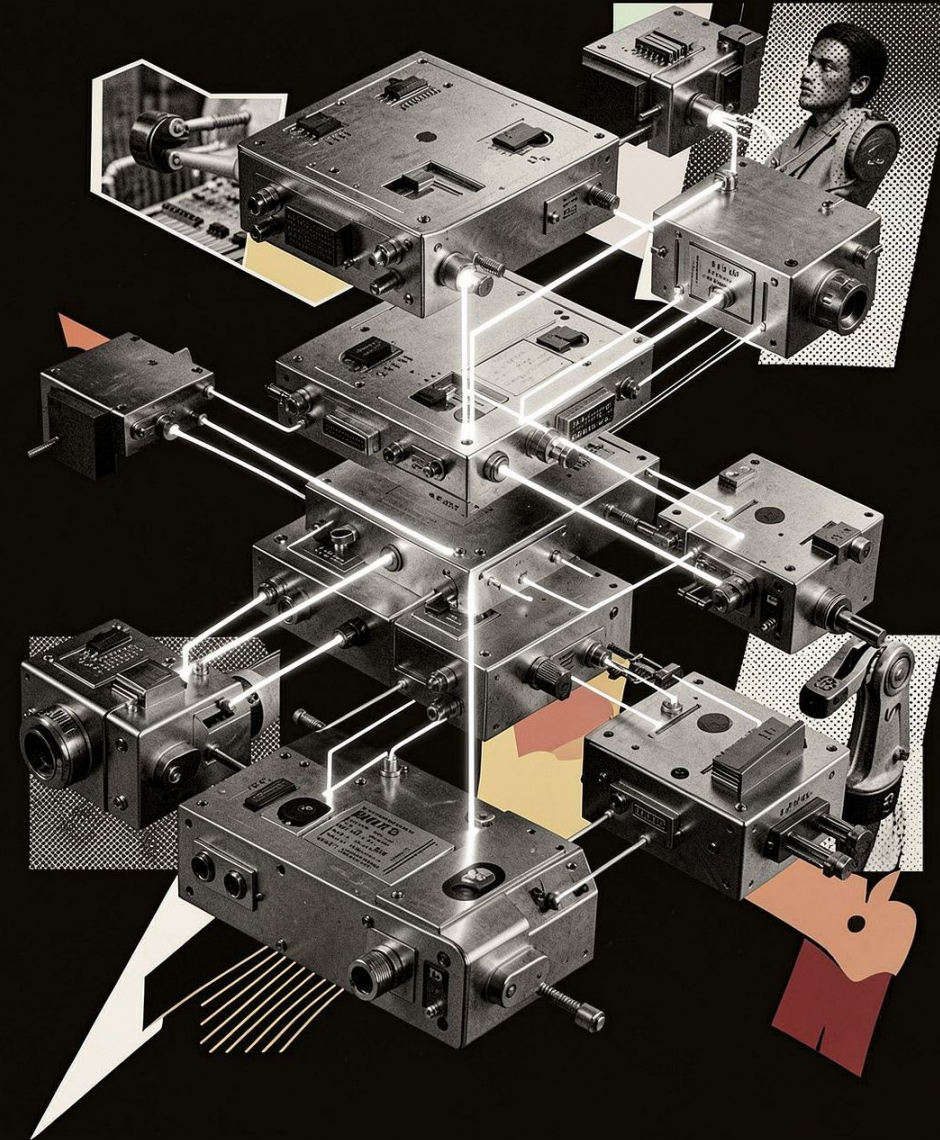
Browser-native structured storage for all financial data; KV for optional non-sensitive edge metadata.

Security

WebAuthn · AES-256-GCM

Hardware-backed biometric authentication and authenticated encryption for all stored and transmitted data.

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Security by Design

Spendly does not trust servers. It does not trust networks. It trusts only the user's device and cryptographic proof. Every architectural decision begins and ends with security.



Zero-Knowledge

No financial data ever leaves the device. The server knows nothing — by design, not by policy.



Offline-First

All core operations — billing, inventory, authentication — function without any internet connection.



WebAuthn

Hardware-backed biometric authentication replaces passwords entirely. No credentials to steal.



AES-256-GCM

All data at rest and in transit is encrypted with authenticated encryption — tamper-evident by default.



Features

Spendly is a complete financial operating layer — not a single-feature app. Every capability is designed to work independently, offline, and without compromise.



Expense Tracking & Analytics

Real-time categorization, spending trends, and savings insights — computed entirely on-device with zero data exposure.



Merchant Billing & QR Invoice

Generate, sign, and share encrypted invoices peer-to-peer. QR-based checkout eliminates hardware dependency.



Inventory & Cash Wallet

Local inventory management with offline stock sync. Digital cash wallet operates without banking infrastructure.



EMI Alerts & Offline Sync

Automated installment reminders and conflict-free peer-to-peer synchronization when connectivity is restored.



Advantages & Future Scope

Spendly is not merely a project — it is a foundation. The current release delivers a production-grade offline-first finance layer. The roadmap extends it into an intelligent, voice-driven, interoperable commerce platform.

Current Advantages

- **Offline-First** — Full functionality without internet
- **Zero Server Cost** — No infrastructure expenses for merchants
- **Privacy by Architecture** — Data never leaves the device
- **Fast P2P Billing** — Instant peer-to-peer invoice exchange

Future Scope

- **AI Insights** — Predictive spending intelligence
- **Voice Billing** — Hands-free merchant interaction
- **UPI & NFC** — Native payment rail integration
- **GST & Multi-Store** — Compliance-ready merchant tools

Spendly

The Future of Private Commerce

Spendly redefines what a financial system can be — one that belongs entirely to its users, operates without dependency, and trusts nothing but cryptographic proof. This is not an incremental improvement. This is a new architecture for commerce.

Offline-First

Always available. Always private.

Zero-Knowledge

Your data. Your device. Your rules.

Peer-to-Peer

No servers. No intermediaries.

Thank you. Questions are welcome.

SPENDLY



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